



Web application for retrospective geomagnetic storm detection, main phase analysis and identification of solar wind influence on storm intensity

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What is the purpose of the app?

- To detect past geomagnetic storms
- To reliably identify their main phases
- To provide flexibility by allowing users to adjust algorithm parameters according to their analysis preferences.
- To identify the solar wind influence on storm intensity

Data input

Hourly values of:

- **Dst** : Disturbance storm time index (nT)
- **IMF**: Interplanetary magnetic field (nT)
- **Bz**: z-component of the IMF (nT)
- **Ey**: Dawn to dusk electric field (mV/m)
- **Speed** of solar wind (km/s)

Time span:
1964-May 2026

Data source:
OMNIWeb

Algorithm

Storm detection
(Total storms)

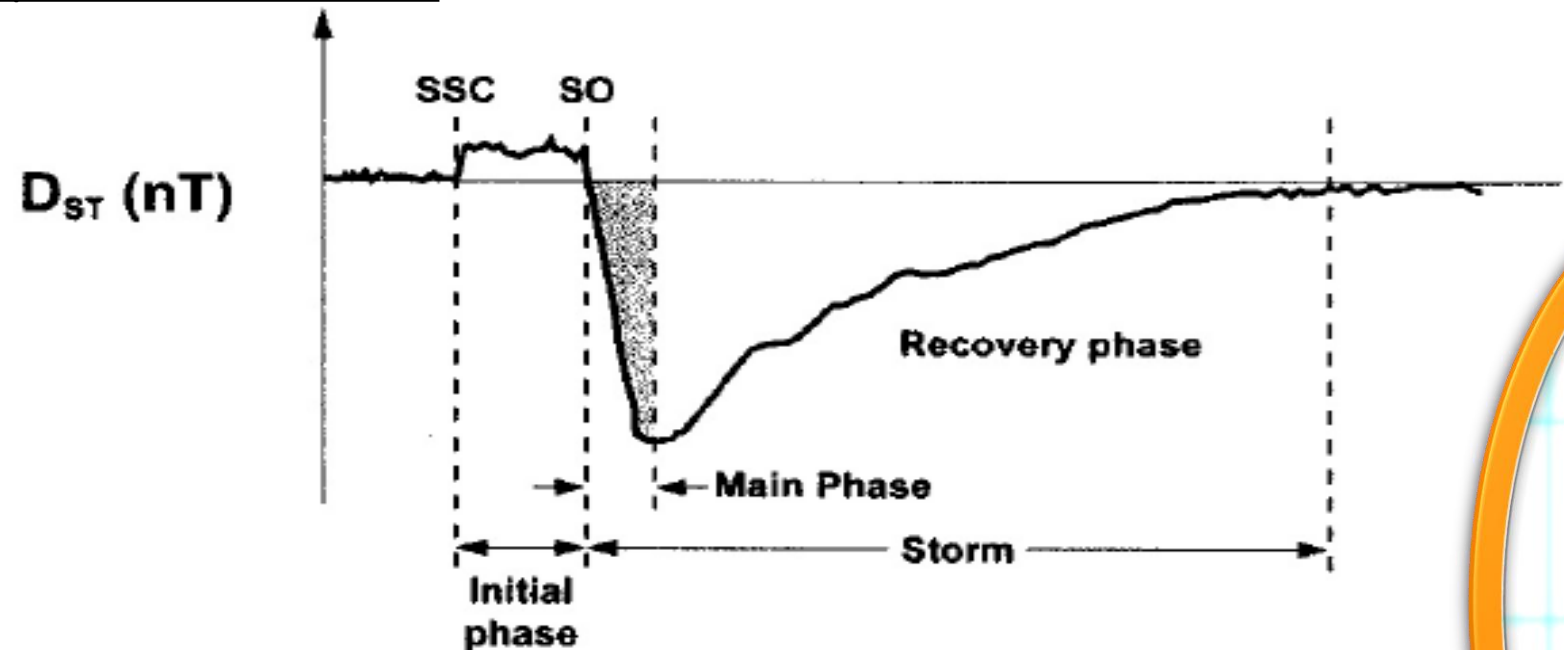
Main phase identification
(Filtered storms)

Solar wind metric computation
(Data-complete, filtered storms)

Correlation analysis
(Solar wind vs |Dst_min|)

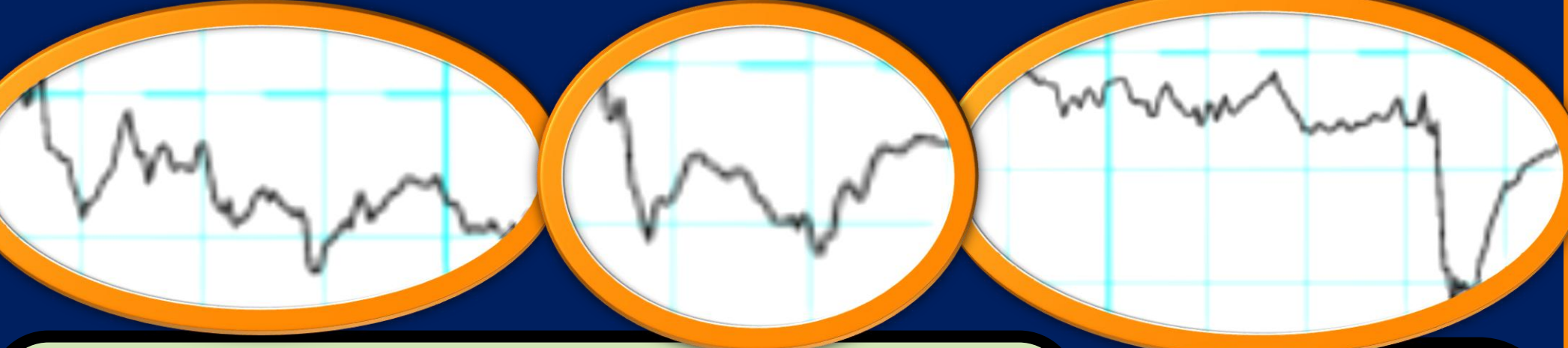
What problems does the app solve?

Typical storm structure:

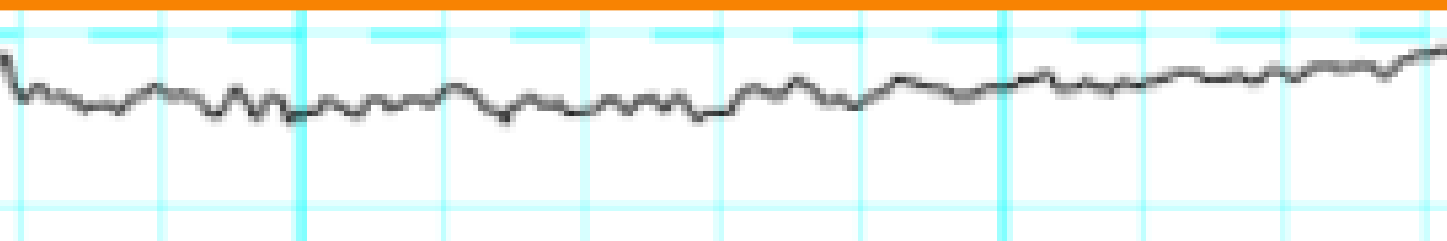


Not all storms follow a well-defined structure like the above. Some events are more ambiguous and may include multiple Dst depressions before the minimum Dst is reached.

- **How is the main phase onset time determined?**
- **When should a separate dip be classified as a distinct storm?**



A Dst depression is classified as a storm once it reaches the -50 nT threshold. However, **low-intensity** Dst depressions may remain below -50 nT for an extended period without developing a clearly defined storm structure, such as a distinct main phase. For this reason, the app allows users to classify such cases as **disturbances** and exclude them from main phase filtering and analysis.



3	999.9	999.9	9999.9	99.99	999.99	-19
4	999.9	999.9	9999.9	99.99	999.99	-29
5	999.9	999.9	9999.9	99.99	999.99	-38
6	999.9	999.9	414.4	99.99	999.99	-39
7	999.9	999.9	414.4	99.99	999.99	-51
8	999.9	999.9	414.4	99.99	999.99	-57
9	999.9	999.9	343.3	99.99	999.99	-68
10	999.9	999.9	343.3	99.99	999.99	-58
11	999.9	999.9	343.3	99.99	999.99	-52
12	999.9	999.9	9999.9	99.99	999.99	-53
13	999.9	999.9	9999.9	99.99	999.99	-51
14	999.9	999.9	9999.9	99.99	999.99	-45
15	999.9	999.9	9999.9	99.99	999.99	-41
16	999.9	999.9	9999.9	99.99	999.99	-46
17	999.9	999.9	9999.9	99.99	999.99	-49
18	999.9	999.9	9999.9	99.99	999.99	-46

OMNI records contain missing values, not all storms are data-complete

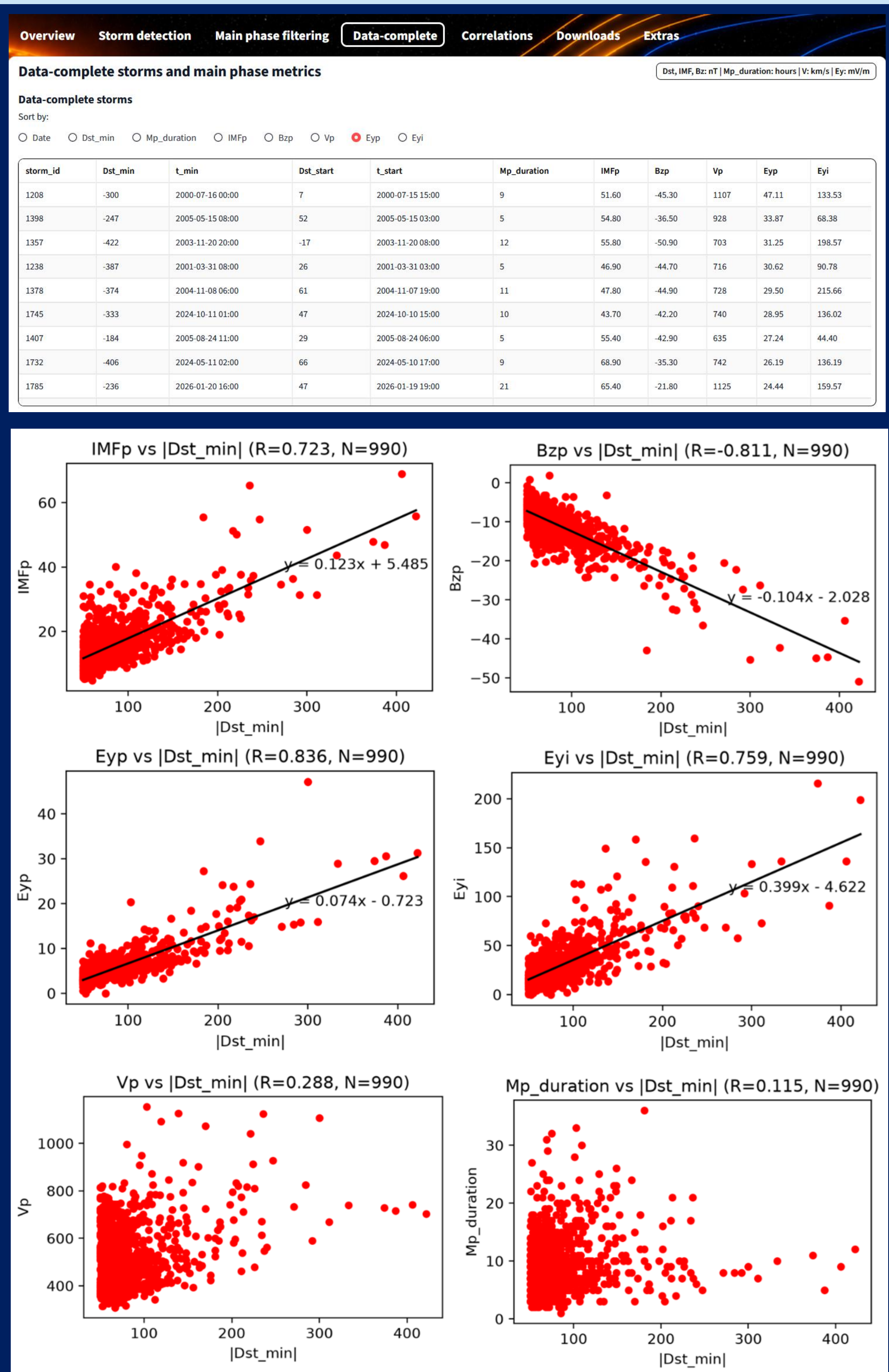
Key outputs

- All detected storms and their metrics, wherever available:
 - Main phase onset time, duration
 - Peak solar wind parameter values during the main phase
 - Eyi (integrated Ey over the main phase)
- Average time delay of minimum Dst from Eyp and Bzp
- Correlation tables and plots
- Exportable CSV files for all results

Result examples

- Total storms: **1796**
 - Moderate (-100 < Dst_min <= -50): **1422**
 - Intense (-250 < Dst_min <= -100): **347**
 - Super-storm (Dst_min <= -250): **27**
- Average main phase duration (h) per class:
M: **9.25**, I: **11.30**, Ss: **13.75**
- Average min Dst delay from Eyp: **3.83 h**
- Average min Dst delay from Bzp: **3.84 h**
- Correlations of metrics with |Dst_min|:
 - Bzp: R= **-0.811**
 - Eyp: R= **0.836**
 - Eyi: R= **0.759**

Web application screenshots



Conclusions

- A web application was developed to detect geomagnetic storms, identify their main phases, and analyze solar wind-Dst relationships.
- User-adjustable parameters allow flexible storm definition and main phase characterization.
- The filtering procedure avoids poorly defined storm intervals and unrealistically long main phase durations.
- IMFp, Bzp, Eyp, and Eyi show strong associations with storm intensity.
- Bzp performs better than total IMFp, as expected due to the role of southward IMF in magnetic reconnection at the magnetopause.
- Eyp and Eyi are also strong predictors of storm intensity, which is reasonable since the dawn-dusk electric field represents the solar wind energy input into the magnetosphere.
- Solar wind speed and main phase duration are weak predictors of storm intensity.